Overfitting and Underfitting

1. **Q:** What happens when a model performs well on training data but poorly on unseen test data?
   A: This is called **Overfitting**.

2. **Q:** What are the loss values like on the training set during overfitting?
   A: Training loss is very low.

3. **Q:** What are the loss values like on the test set during overfitting?
   A: Test loss is high.

4. **Q:** What does the model learn in overfitting instead of the true data pattern?
   A: It learns **noise or random fluctuations**.

5. **Q:** Give an example of an overfitting model type.
   A: A **high-degree polynomial model** for simple data.

6. **Q:** When does underfitting occur?
   A: When the model fails to capture the **underlying data pattern**.

7. **Q:** What are the loss values like in underfitting on both datasets?
   A: Both **training and test losses are high.**

8. **Q:** What kind of R^2 value indicates underfitting?
   A: Low R^2 value.

9. **Q:** What phrase refers to an underfitted model?
   A: "Model too simple."

10. **Q:** What phrase refers to an overfitted model?
    A: "Model too complex."

11. **Q:** What characterizes an optimally selected model?
    A: Low training loss and low test loss.

Methods for Dealing with Overfitting

12. **Q:** Name three methods used to reduce overfitting.
    A: Ridge Regression, **Lasso Regression**, and **Elastic Net.**

13. **Q:** What is the goal of the penalty term in Ridge Regression?
    A: To **control model complexity**.

14. **Q:** The Ridge penalty is applied to which model parameters?
    A: The **magnitude of regression coefficients.**

Ridge Regression Loss Function and Formulation

15. **Q:** What is the λ (lambda) term in Ridge Regression called?
    A: Regularization parameter.

16. **Q:** What is the expression (\sum_{i=1}^{n}(y_i - (\beta_0 + \sum_{j=1}^{n}x_{ij}\beta_j))^2)?
    A: The **Residual Sum of Squares (RSS).**

17. **Q:** Which term represents the penalty in Ridge Regression?
    A: (\lambda \sum_{j=1}^{p}\beta_j^2).

18. **Q:** What does the penalty term prevent weights from becoming?
    A: Too large.

19. **Q:** What does Ridge Regression minimize overall?
    A: Sum of squared errors plus penalty term.

20. **Q:** What constraint is imposed on the coefficients?
    A: The **sum of squared coefficients ≤ constant.**

21. **Q:** What is the matrix form of the Ridge Loss Function?
    A: (L(\beta) = (Y - X\beta)^T(Y - X\beta) + \lambda \beta^T\beta.)**

Derivation and Solution of Ridge Regression

22. **Q:** What is obtained by setting the derivative of the Ridge loss to zero?
    A: ((X^TX + \lambda I)\beta = X^TY.)**

23. **Q:** What is the Ridge Regression estimator formula?
    A: (\hat{\beta}_{ridge} = (X^TX + \lambda I)^{-1}X^TY.)**

24. **Q:** When does Ridge Regression reduce to OLS?
    A: When **$\lambda = 0$.**

25. **Q:** What is $((X^TX + \lambda I)^{-1})$ part of?
    A: The **Ridge Regression coefficient solution.**

Advantages of Ridge Regression

26. **Q:** What does Ridge Regression reduce in terms of model dependency?
    A: Dependency on specific parameters.

27. **Q:** What model property does Ridge Regression improve?
    A: Generalization ability.

28. **Q:** What does the λ parameter balance?
    A: Bias and variance.

29. **Q:** When does OLS fail to produce a solution?
    A: When **(X^TX)** is **singular (non-invertible).**

30. **Q:** How does regularization help when (X^TX) is singular?
    A: Adds **λI**, making the matrix invertible and stable.